Kruskal's Algorithm:

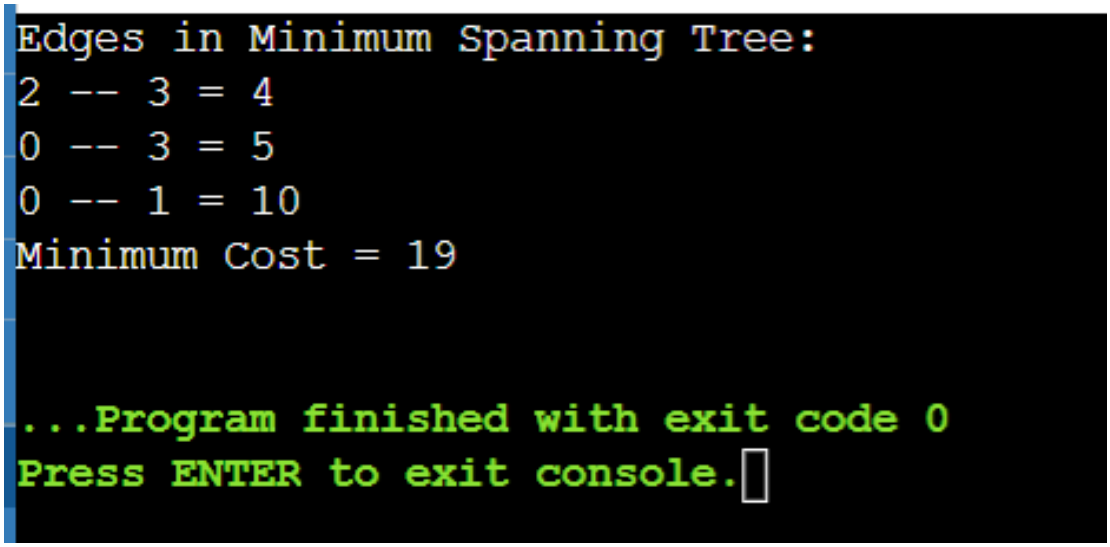
Definition:

Kruskal's Algorithm is a greedy algorithm used to find the Minimum Spanning Tree (MST) of a connected, weighted, undirected graph. It selects the edges with the smallest weights without forming cycles.

Algorithm:

1. Start with a graph containing vertices and weighted edges.
2. Sort all edges in ascending order of weight.
3. Select the edge with the smallest weight.
4. Check whether adding the edge creates a cycle.
5. If no cycle is formed, add the edge to the MST.
6. Repeat until \(V-1\) edges are selected.
7. Calculate the total cost of the MST.

Out put:



Time Complexity

O(Elog E)

Sorting the edges takes O(Elog E) time.

Space Complexity

(O(V+E)